

Class 20 Primes

Goals

1. Write $\gcd(a,b)$ as a linear combination of a and b .
2. State and prove Euclid's Lemma.
3. Know what the Fundamental Theorem of Arithmetic says.
4. Prove there are infinitely many primes.

We saw last class how using the division algorithm repeatedly we can determine the $\gcd(a,b)$. This method was called the Euclidean Algorithm.

By reversing the Euclidean Algorithm, it is always possible to write the $\gcd(a,b)$ as an integer linear combination of a and b . In fact the $\gcd(a,b)$ is the smallest positive integer that can be written as a linear combination of a and b .

Let $d = \gcd(a,b)$.

Theorem: If n is any common divisor of a and b , then n will divide the $\gcd(a,b)$.

What does this mean if a and b are relatively prime?

Fundamental Theorem of Arithmetic

Every integer $m > 1$ may be written as a product of primes to powers

This factorization is unique up to the order in which the primes are written.

Euclid's Lemma: For prime, p , and integers a and b , if $p|ab$, then $p|a$ or $p|b$.

Proof:

Theorem: There are infinitely many primes.

Proof by contradiction